

This document provides additional assistance with wiring your Extron IP Link Pro Control Processor to your device. Different components may require a different wiring scheme than those listed below.

For complete operating instructions, refer to the user's manual for the specific IP Link Pro Control Processor or the documentation supplied by the manufacturer of the controlled device.

For more information on using Global Scriptor Modules, refer to the "[Guide to Using Scriptor Modules](#)" document.

Device Specifications

Device Type: Video Projector
Manufacturer: Christie
Firmware Version: N/A
Model(s): LX400

Tested on the Following Software and Firmware Versions

IP Link Pro Control Processor Firmware	Global Scriptor Version
3.03.0000-b006	2.3.0

Version History

Module Version	Date	Notes
1_0_0_0	6/4/2019	Initial Version

Module Notes

- Unidirectional variable must be set to 'True' if status is not required. Default value is 'False'.
Example: `InterfaceName.Unidirectional = 'True'`
- connectionCounter variable must be set to the number of queries that will be sent to the device before displaying 'Disconnected' if no response is received. Default value is 15.
Example: `InterfaceName.connectionCounter = 5`
- Ethernet only:
If login credentials are required, devicePassword must be set accordingly.
Example: `InterfaceName.devicePassword = 'extron'`

Supported Classes and Examples

SerialClass
<code>InterfaceName = ModuleName.SerialClass(ProcessorName, 'COM1', Model='LX400')</code>
SerialOverEthernetClass
<code>InterfaceName = ModuleName.SerialOverEthernetClass('192.168.254.254', 2001, Model='LX400')</code>
EthernetClass
<code>InterfaceName = ModuleName.EthernetClass('192.168.254.254', 23, Model='LX400')</code>

Control Commands

Format with Qualifier:

```
InterfaceName.Set(Command, Value, {'Qualifier Key': 'Qualifier Value'})
```

Format with Qualifier:

```
InterfaceName.Set(Command, Value)
```

Command AspectRatio	Value '4:3' 'Small'	Value '16:9' 'Normal'	Value '14:9'
# AspectRatio example InterfaceName.Set('AspectRatio', '4:3')			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Set('AudioMute', 'On')			
Command AutoImage	Value None		
# AutoImage example InterfaceName.Set('AutoImage', None)			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Set('Freeze', 'On')			
Command Input	Value 'RGB 1' 'Video'	Value 'RGB 2' 'S-Video'	Value 'HDMI' 'Component'
# Input example InterfaceName.Set('Input', 'RGB 1')			
Command Power	Value 'On'	Value 'Off'	
# Power example InterfaceName.Set('Power', 'On')			
Command VideoMute	Value 'On'	Value 'Off'	
# VideoMute example InterfaceName.Set('VideoMute', 'On')			
Command Volume	Value 'Up'	Value 'Down'	
Qualifier Key 'Input'	Qualifier Value 'RGB 1' 'Video'	Qualifier Value 'RGB 2' 'S-Video'	Qualifier Value 'HDMI' 'Component'
# Volume example InterfaceName.Set('Volume', 'Up', {'Input': 'RGB 1'})			

Status Available

For all commands, call Update to receive the latest status. ConnectionStatus does not support the Update function and is triggered by the device providing a successful response to other Update function calls.

Format with Qualifier:

```
InterfaceName.Update(Command, {'Qualifier Key': 'Qualifier Value'})
Value = InterfaceName.ReadStatus(Command, {'Qualifier Key': 'Qualifier Value'})
InterfaceName.SubscribeStatus(Command, {'Qualifier Key': 'Qualifier Value'},
```

FeedbackHandler)

FeedbackHandler will be called only when the specified qualifier gets a new status.

Format without Qualifier:

```
InterfaceName.Update(Command)
Value = InterfaceName.ReadStatus(Command)
InterfaceName.SubscribeStatus(Command, None, FeedbackHandler)
FeedbackHandler will be called when any qualifier gets a new status.
```

Command AspectRatio	Value '4:3' 'Small'	Value '16:9' 'Normal'	Value '14:9'
# AspectRatio example InterfaceName.Update('AspectRatio') Value = InterfaceName.ReadStatus('AspectRatio') InterfaceName.SubscribeStatus('AspectRatio', None, FeedbackHandler)			
Command AudioMute	Value 'On'	Value 'Off'	
# AudioMute example InterfaceName.Update('AudioMute') Value = InterfaceName.ReadStatus('AudioMute') InterfaceName.SubscribeStatus('AudioMute', None, FeedbackHandler)			
Command ConnectionStatus	Value 'Connected'	Value 'Disconnected'	
Value = InterfaceName.ReadStatus('ConnectionStatus') InterfaceName.SubscribeStatus('ConnectionStatus', None, FeedbackHandler)			
Command DeviceStatus	Value 'Normal' 'Lamp Error' 'Lamp Time Error'	Value 'Cover Error' 'Temp Error' 'Cold Error'	Value 'Fan Error' 'Air Flow Error' 'Filter Error'
# DeviceStatus example InterfaceName.Update('DeviceStatus') Value = InterfaceName.ReadStatus('DeviceStatus') InterfaceName.SubscribeStatus('DeviceStatus', None, FeedbackHandler)			
Command FilterUsage	Value Hours		
# FilterUsage example InterfaceName.Update('FilterUsage') Value = InterfaceName.ReadStatus('FilterUsage') InterfaceName.SubscribeStatus('FilterUsage', None, FeedbackHandler)			
Command Freeze	Value 'On'	Value 'Off'	
# Freeze example InterfaceName.Update('Freeze') Value = InterfaceName.ReadStatus('Freeze') InterfaceName.SubscribeStatus('Freeze', None, FeedbackHandler)			
Command Input	Value 'RGB 1'	Value 'RGB 2'	Value 'HDMI'

Global Scriptor Module Communication Sheet

	'Video'	'S-Video'	'Component'
# Input example InterfaceName.Update('Input') Value = InterfaceName.ReadStatus('Input') InterfaceName.SubscribeStatus('Input', None, FeedbackHandler)			
Command LampUsage	Value Hours		
# LampUsage example InterfaceName.Update('LampUsage') Value = InterfaceName.ReadStatus('LampUsage') InterfaceName.SubscribeStatus('LampUsage', None, FeedbackHandler)			
Command Power	Value 'On'	Value 'Off'	Value 'Cooling Down'
# Power example InterfaceName.Update('Power') Value = InterfaceName.ReadStatus('Power') InterfaceName.SubscribeStatus('Power', None, FeedbackHandler)			
Command VideoMute	Value 'On'	Value 'Off'	
# VideoMute example InterfaceName.Update('VideoMute') Value = InterfaceName.ReadStatus('VideoMute') InterfaceName.SubscribeStatus('VideoMute', None, FeedbackHandler)			
Command VolumeStatus	Value —		
Qualifier Key 'Input'	Qualifier Value 'RGB 1' 'Video'	Qualifier Value 'RGB 2' 'S-Video'	Qualifier Value 'HDMI' 'Component'
# VolumeStatus example InterfaceName.Update('VolumeStatus', {'Input': 'RGB 1'}) Value = InterfaceName.ReadStatus('VolumeStatus', {'Input': 'RGB 1'}) InterfaceName.SubscribeStatus('VolumeStatus', None, FeedbackHandler)			

Cable and Adapter Requirements

Captive Screw to Female DB9 RS-232 Serial Cable

Notes for the Device

Serial communication

Port Type: RS-232

Baud Rate: 19200

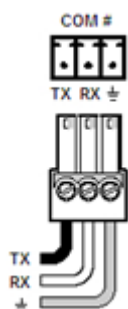
Data Bits: 8

Parity: None

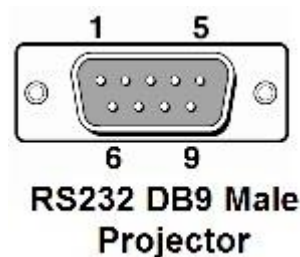
Stop Bits: One

Flow Control: None

Pin Assignments Diagram



Signal	Main Cable	Pin	Signal
TxD	→	2	RxD
RxD	←	3	TxD
GND	→	5	GND



Network communication

When configuring the Ethernet module, be sure device settings match those of the Global Scriptor ethernet interface.

Port Type: Ethernet

Default Port: 23

Logon Credentials Supported: Yes

Default Password:

Multi-Connection Undetermined

Capabilities:

Port Changeability: Yes

Ethernet Module Configuration Description

Please refer to user manual for settings and changes to the network communication

Notes for the Device

Appendix A. Set Commands

Aspect Ratio 14:9	\xBE\xEF\x03\x06\x00\xCE\xD6\x01\x00\x08 \x09\x00
Aspect Ratio 16:9	\xBE\xEF\x03\x06\x00\x0E\xD1\x01\x00\x08 \x01\x00
Aspect Ratio 4:3	\xBE\xEF\x03\x06\x00\x9E\xD0\x01\x00\x08 \x00\x00
Aspect Ratio Normal	\xBE\xEF\x03\x06\x00^\xDD\x01\x00\x08 \x10\x00
Aspect Ratio Small	\xBE\xEF\x03\x06\x00\xFE\xD1\x01\x00\x08 \x02\x00
Audio Mute Off	\xBE\xEF\x03\x06\x00F\xD3\x01\x00\x02 \x00\x00
Audio Mute On	\xBE\xEF\x03\x06\x00\xD6\xD2\x01\x00\x02 \x01\x00
Auto Image None	\xBE\xEF\x03\x06\x00\x91\xD0\x06\x00\x0A \x00\x00
Freeze Off	\xBE\xEF\x03\x06\x00\x83\xD2\x01\x00\x020\x00\x00
Freeze On	\xBE\xEF\x03\x06\x00\x13\xD3\x01\x00\x020\x01\x00
Input Component	\xBE\xEF\x03\x06\x00\xAE\xD1\x01\x00\x00 \x05\x00
Input HDMI	\xBE\xEF\x03\x06\x00\x0E\xD2\x01\x00\x00 \x03\x00
Input RGB 1	\xBE\xEF\x03\x06\x00\xFE\xD2\x01\x00\x00 \x00\x00
Input RGB 2	\xBE\xEF\x03\x06\x00>\xD0\x01\x00\x00 \x04\x00
Input S-Video	\xBE\xEF\x03\x06\x00\x9E\xD3\x01\x00\x00 \x02\x00
Input Video	\xBE\xEF\x03\x06\x00n\xD3\x01\x00\x00 \x01\x00
Power Off	\xBE\xEF\x03\x06\x00*\xD3\x01\x00\x00` \x00\x00
Power On	\xBE\xEF\x03\x06\x00\xBA\xD2\x01\x00\x00` \x01\x00
Video Mute Off	\xBE\xEF\x03\x06\x00\xFB\xD8\x01\x00 0\x00\x00
Video Mute On	\xBE\xEF\x03\x06\x00k\xD9\x01\x00 0\x01\x00
Volume Down Input Component	\xBE\xEF\x03\x06\x00\xB6\xCD\x05\x00e \x00\x00
Volume Down Input HDMI	\xBE\xEF\x03\x06\x00>\xCD\x05\x00c \x00\x00
Volume Down Input RGB 1	\xBE\xEF\x03\x06\x00z\xCD\x05\x00` \x00\x00
Volume Down Input RGB 2	\xBE\xEF\x03\x06\x00J\xCC\x05\x00d \x00\x00
Volume Down Input S-Video	\xBE\xEF\x03\x06\x00\xC2\xCC\x05\x00b \x00\x00
Volume Down Input Video	\xBE\xEF\x03\x06\x00\x86\xCC\x05\x00a \x00\x00
Volume Up Input Component	\xBE\xEF\x03\x06\x00g\xCC\x04\x00e \x00\x00
Volume Up Input HDMI	\xBE\xEF\x03\x06\x00\xEF\xCC\x04\x00c \x00\x00
Volume Up Input RGB 1	\xBE\xEF\x03\x06\x00\xAB\xCC\x04\x00` \x00\x00
Volume Up Input RGB 2	\xBE\xEF\x03\x06\x00\x9B\xCD\x04\x00d \x00\x00
Volume Up Input S-Video	\xBE\xEF\x03\x06\x00\x13\xCD\x04\x00b \x00\x00
Volume Up Input Video	\xBE\xEF\x03\x06\x00W\xCD\x04\x00a \x00\x00

Appendix B. Update Commands

Aspect Ratio	\xBE\xEF\x03\x06\x00\xAD\xD0\x02\x00\x08 \x00\x00
Audio Mute	\xBE\xEF\x03\x06\x00u\xD3\x02\x00\x02 \x00\x00
Device Status	\xBE\xEF\x03\x06\x00\xD9\xD8\x02\x00 ` \x00\x00
Filter Usage	\xBE\xEF\x03\x06\x00\xC2\xF0\x02\x00\xA0\x10\x00\x00
Freeze	\xBE\xEF\x03\x06\x00\xB0\xD2\x02\x00\x020\x00\x00
Input	\xBE\xEF\x03\x06\x00\xCD\xD2\x02\x00\x00 \x00\x00
Lamp Usage	\xBE\xEF\x03\x06\x00\xC2\xFF\x02\x00\x90\x10\x00\x00
Power	\xBE\xEF\x03\x06\x00\x19\xD3\x02\x00\x00` \x00\x00
Video Mute	\xBE\xEF\x03\x06\x00\xC8\xD8\x02\x00 0\x00\x00
Volume Status Input Component	\xBE\xEF\x03\x06\x00\x01\xCC\x02\x00e \x00\x00
Volume Status Input HDMI	\xBE\xEF\x03\x06\x00\x89\xCC\x02\x00c \x00\x00
Volume Status Input RGB 1	\xBE\xEF\x03\x06\x00\xCD\xCC\x02\x00` \x00\x00
Volume Status Input RGB 2	\xBE\xEF\x03\x06\x00\xFD\xCD\x02\x00d \x00\x00
Volume Status Input S-Video	\xBE\xEF\x03\x06\x00u\xCD\x02\x00b \x00\x00
Volume Status Input Video	\xBE\xEF\x03\x06\x001\xCD\x02\x00a \x00\x00